Solutions to MP363 Problem Set 2

P.1

For this problem, we want to show that each of the three inner products actually satisfies the necessary requirements. They obviously do, but it's always good to double-check...

(a) If $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ and $w = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$, $v+w$ is $\begin{pmatrix} v_1+w_1 \\ \vdots \\ v_n+w_n \end{pmatrix}$, so the ith component is $(v+w)_i = v_i + w_i$. Thus,

(i) $\langle u|v+w\rangle = \sum_i u_i^*(v_i+w_i) = \sum_i u_i^* v_i + \sum_i u_i^* w_i = \langle u|v\rangle + \langle u|w\rangle$.

The same idea applies for $\langle u+v|w\rangle$:

(ii) $\langle u+v|w\rangle = \sum_i (u_i+v_i)^* w_i = \sum_i u_i^* w_i + \sum_i v_i^* w_i = \langle u|w\rangle + \langle v|w\rangle$.

If α is a const, then $\alpha v = \begin{pmatrix} \alpha v_1 \\ \alpha v_n \end{pmatrix}$ i.e. $(\alpha v)_i = \alpha v_i$. Thus,

(iii) $\langle u|\alpha v\rangle = \sum_i u_i^*(\alpha v_i) = \alpha \sum_i u_i^* v_i = \alpha\langle u|v\rangle$.

Similarly,

(iv) $\langle \alpha u|v\rangle = \sum_i (\alpha u_i)^* v_i = \sum_i \alpha^* u_i^* v_i = \alpha^*\langle u|v\rangle$

Finally, if we complex-conjugate $\langle u|v\rangle$, we see that

(v) $\langle u|v\rangle^* = \left(\sum_i u_i^* v_i\right)^* = \sum_i (u_i^* v_i)^* = \sum_i u_i v_i^* = \sum_i v_i^* u_i = \langle v|u\rangle$

and so $\langle u|v\rangle = \sum_i u_i^* v_i = u^\dagger \cdot v$ is indeed an inner product

(b) If two functions are added the resulting function is just the sum of the two: $(v+w)(x) = v(x)+w(x)$. Thus,

(i) $\langle u|v+w\rangle = \int_{-\infty}^{\infty} u^*(x)(v(x)+w(x))\,dx = \int_{-\infty}^{\infty} u^*(x)v(x)\,dx + \int_{-\infty}^{\infty} u^*(x)w(x)\,dx$
$= \langle u|v\rangle + \langle u|w\rangle$

and

(ii) $\langle u+v|w\rangle = \int_{-\infty}^{\infty} (u(x)+v(x))^* w(x)\,dx = \int_{-\infty}^{\infty} (u^*(x)+v^*(x))\,w(x)\,dx$
$= \int_{-\infty}^{\infty} u^*(x)w(x)\,dx + \int_{-\infty}^{\infty} v^*(x)w(x)\,dx = \langle u|w\rangle + \langle v|w\rangle$.

$(\alpha v)(x)$ is the function $\alpha v(x)$, so

(iii) $\langle u|\alpha v\rangle = \int_{-\infty}^{\infty} u^*(x)(\alpha v(x))\,dx = \alpha\int_{-\infty}^{\infty} u^*(x)v(x)\,dx$
$= \alpha\langle u|v\rangle$

and

(iv) $\langle \alpha u|v\rangle = \int_{-\infty}^{\infty} (\alpha u(x))^* v(x)\,dx = \int_{-\infty}^{\infty} \alpha^* u^*(x)v(x)\,dx$
$= \alpha^*\langle u|v\rangle$.

Finally,

$\langle u|v\rangle^* = \left(\int_{-\infty}^{\infty} u^*(x)v(x)\,dx\right)^* = \int_{-\infty}^{\infty} (u^*(x)v(x))^*\,dx^*$

$$= \int_{-\infty}^{\infty} u(x) v^*(x)\,dx = \int_{-\infty}^{\infty} v^*(x) u(x)\,dx = \langle v|u\rangle$$

because x is a real variable $(x^* = x)$.

Thus, $\boxed{\text{both of these examples define inner products}}$ which is good.

(b) There are a number of ways to prove all these properties of the adjoint operation, but here I'll do it by looking at the individual elements in the matrices. In other words, I'll use $(A^T)_{ij} = A_{ji}$ rather than just A^T. This means that the def'n of the adjoint gives

$$(A^\dagger)_{ij} = (A_{ji})^*$$

so on we go.

(a) $(A+B)_{ij} = A_{ij} + B_{ij}$, so $\left[(A+B)^\dagger\right]_{ij} = (A+B)^*_{ji}$

$$= (A_{ji} + B_{ji})^* = (A_{ji})^* + (B_{ji})^* = (A^\dagger)_{ij} + (B^\dagger)_{ij} = (A^\dagger + B^\dagger)_{ij}$$

and so $\boxed{(A+B)^\dagger = A^\dagger + B^\dagger}$.

(b) αA is the matrix with elements $(\alpha A)_{ij} = \alpha A_{ij}$, so

$$(\alpha A^\dagger)_{ij} = (\alpha A)^*_{ji} = (\alpha A_{ji})^* = \alpha^* A^*_{ji} = \alpha^* (A^\dagger)_{ij}$$
$$= (\alpha^* A^\dagger)_{ij}$$

and so $\boxed{(\alpha A)^\dagger = \alpha^* A^\dagger}$

(c) $(AB)_{ij} = \sum_k A_{ik} B_{kj}$, because that's the def'n of matrix multiplication. Thus,

$$\left[(AB)^\dagger\right]_{ij} = \left[(AB)_{ji}\right]^* = \left[\sum_k A_{jk} B_{ki}\right]^*$$

$$= \sum_k (A_{jk})^* (B_{ki})^* = \sum_k (A^\dagger)_{kj} (B^\dagger)_{ik}$$

$$= \sum_k (B^\dagger)_{ik} (A^\dagger)_{kj} = (B^\dagger A^\dagger)_{ij}$$

so $\boxed{(AB)^\dagger = B^\dagger A^\dagger.}$

(d) Finally, the fact that two adjoint operations return us to the original matrix:

$$[(A^\dagger)^\dagger]_{ij} = [(A^\dagger)_{ji}]^* = [(A_{ij})^\dagger]^* = A_{ij}$$

so $\boxed{(A^\dagger)^\dagger = A}$

As stated, these hold for any of the operators we've talked about, not just matrices. So regardless of if A is a matrix, a differential operator, or whatever, $(\alpha A)^\dagger = \alpha^* A^\dagger$ and so on. Thus, for example, $(XP)^\dagger = P^\dagger X^\dagger = PX$ even when X & P are acting on functions rather than column vectors. Handy stuff to know.

P.3

The point of this problem is to illustrate the idea of the orthonormality of functions. Orthonormal vectors in $\mathbb{R}^3$ are well-known: $\hat{i}\cdot\hat{i} = 1$ and $\hat{j}\cdot\hat{k} = 0$, for example. But the concept extends to all types of vectors if we have an inner product.

So what we need to do is find the six inner products $\langle\phi_0|\phi_0\rangle$, $\langle\phi_1|\phi_1\rangle$, $\langle\phi_2|\phi_2\rangle$, $\langle\phi_0|\phi_1\rangle$, $\langle\phi_0|\phi_2\rangle$ and $\langle\phi_1|\phi_2\rangle$. The first three should be 1 and the last three should be zero. As stated, this will be sufficient to confirm that $\langle\phi_i|\phi_j\rangle = \delta_{ij}$. To do these, we'll need the following trig identities (which all of you should know by heart by now):

$\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$, $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$, and

$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$.

So let's go:

$$\langle\phi_0|\phi_0\rangle = \int_{-L}^{L} \phi_0(x)^* \phi_0(x)\,dx = \int_{-L}^{L} \frac{1}{\sqrt{2L}}\cdot\frac{1}{\sqrt{2L}}\,dx = \frac{1}{2L}\int_{-L}^{L} dx$$

$$= \frac{1}{2L}\left[x\right|_{-L}^{L} = \frac{1}{2L}\left[L-(-L)\right] = 1 \quad ✓$$

$$\langle\phi_1|\phi_1\rangle = \int_{-L}^{L} \phi_1(x)^* \phi_1(x)\,dx = \int_{-L}^{L} \frac{1}{L}\sin^2\left(\frac{\pi x}{L}\right)$$

$$= \frac{1}{2L}\int_{-L}^{L}\left[1-\cos\left(\frac{2\pi x}{L}\right)\right]dx = \frac{1}{2L}\left[x - \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right|_{-L}^{L}$$

4

$$= \frac{1}{2L}\left[\left(L - \frac{L}{2\pi}\sin(2\pi)\right) - \left(-L - \frac{L}{2\pi}\sin(-2\pi)\right)\right] = 1$$ ✓

$$\langle\phi_2|\phi_2\rangle = \int_{-L}^{L} \phi_2(x)^* \phi_2(x)\,dx = \int_{-L}^{L} \frac{1}{L}\cos^2\left(\frac{\pi x}{L}\right) dx$$

$$= \frac{1}{2L}\int_{-L}^{L}\left[1+\cos\left(\frac{2\pi x}{L}\right)\right]dx = \frac{1}{2L}\left[x + \frac{L}{2\pi}\sin\left(\frac{2\pi x}{L}\right)\right]_{-L}^{L}$$

$$= \frac{1}{2L}\left[\left(L + \frac{L}{2\pi}\sin(2\pi)\right) - \left(-L + \frac{L}{2\pi}\sin(-2\pi)\right)\right] = 1$$ ✓

$$\langle\phi_0|\phi_1\rangle = \int_{-L}^{L} \phi_0(x)^* \phi_1(x)\,dx = \int_{-L}^{L} \frac{1}{\sqrt{2}L}\sin\left(\frac{\pi x}{L}\right) dx$$

$$= \frac{1}{\sqrt{2}L}\left[-\frac{L}{\pi}\cos\left(\frac{\pi x}{L}\right)\right]_{-L}^{L} = \frac{1}{\sqrt{2}L}\left[-\frac{L}{\pi}\cos(\pi) + \frac{L}{\pi}\cos(-\pi)\right] = 0$$ ✓

$$\langle\phi_0|\phi_2\rangle = \int_{-L}^{L} \phi_0(x)^* \phi_2(x)\,dx = \int_{-L}^{L} \frac{1}{\sqrt{2}L}\cos\left(\frac{\pi x}{L}\right) dx$$

$$= \frac{1}{\sqrt{2}L}\left[\frac{L}{\pi}\sin\left(\frac{\pi x}{L}\right)\right]_{-L}^{L} = \frac{1}{\sqrt{2}L}\left[\frac{L}{\pi}\sin(\pi) - \frac{L}{\pi}\sin(-\pi)\right] = 0$$ ✓

$$\langle\phi_1|\phi_2\rangle = \int_{-L}^{L} \phi_1(x)^* \phi_2(x)\,dx = \int_{-L}^{L} \frac{1}{L}\sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right) dx$$

$$= \frac{1}{2L}\int_{-L}^{L}\sin\left(\frac{2\pi x}{L}\right)dx = \frac{1}{2L}\left[-\frac{L}{2\pi}\cos\left(\frac{2\pi x}{L}\right)\right]_{-L}^{L}$$

$$= \frac{1}{2L}\left[-\frac{L}{2\pi}\cos(2\pi) + \frac{L}{2\pi}\cos(-2\pi)\right] = 0$$ ✓

so we see that $\boxed{\langle\phi_i|\phi_j\rangle = \delta_{ij}}$ is, in fact, correct.

P.4

We already know that we'll get $\langle\psi|H\zeta\rangle - \langle\zeta|H\psi\rangle = 0$, because $H^\dagger = H$ was already shown. But this version of the proof is useful because it gives the general method of showing self-adjointness (or lack thereof) of differential operators, of which there are lots. So let's do it.

(a) This is easy, since it follows immediately from the product rule for derivatives:

$$\frac{\partial}{\partial x}\left[f\frac{\partial g}{\partial x} - \frac{\partial f}{\partial x}g\right] = \frac{\partial f}{\partial x}\frac{\partial g}{\partial x} + f\frac{\partial^2 g}{\partial x^2} - \frac{\partial^2 f}{\partial x^2}g - \frac{\partial f}{\partial x}\frac{\partial g}{\partial x}$$

$$= \boxed{f\frac{\partial^2 g}{\partial x^2} - \frac{\partial^2 f}{\partial x^2} g.}$$

That's it, basically.

(b) Now to use it: the two inner products are

$$\langle \psi | H \zeta \rangle = \int_{-\infty}^{\infty} \psi^* \left(-\frac{\hbar^2}{2m}\frac{\partial^2 \zeta}{\partial x^2} + V\zeta \right) dx$$

$$= -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \psi^* \frac{\partial^2 \zeta}{\partial x^2} dx + \int_{-\infty}^{\infty} V\psi^* \zeta \, dx$$

and

$$\langle H\psi | \zeta \rangle = \int_{-\infty}^{\infty} \left(-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} + V\psi \right)^* \zeta \, dx$$

$$= \int_{-\infty}^{\infty} \left(-\frac{\hbar^2}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + V\psi^* \right) \zeta \, dx$$

$$= -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} \zeta + \int_{-\infty}^{\infty} V\psi^* \zeta \, dx$$

because V is real, as stated. Note that the second term in each is the same, and so will cancel out when we subtract the second from the first. Thus, we have

$$\langle \psi | H \zeta \rangle - \langle H\psi | \zeta \rangle = -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \psi^* \frac{\partial^2 \zeta}{\partial x^2} dx + \frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \frac{\partial^2 \psi^*}{\partial x^2} \zeta \, dx$$

$$= -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \left(\psi^* \frac{\partial^2 \zeta}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} \zeta \right) dx$$

But using the result from (a), $\psi^* \frac{\partial^2 \zeta}{\partial x^2} - \frac{\partial^2 \psi^*}{\partial x^2} \zeta = \frac{\partial}{\partial x}\left(\psi^* \frac{\partial \zeta}{\partial x} - \frac{\partial \psi^*}{\partial x} \zeta \right)$. Thus,

$$\langle \psi | H \zeta \rangle - \langle H\psi | \zeta \rangle = -\frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \frac{\partial}{\partial x}\left(\psi^* \frac{\partial \zeta}{\partial x} - \frac{\partial \psi^*}{\partial x} \zeta \right) dx$$

$$= -\frac{\hbar^2}{2m} \left[\psi^* \frac{\partial \zeta}{\partial x} - \frac{\partial \psi^*}{\partial x} \zeta \right]_{-\infty}^{\infty}$$

from the Fundamental theorem of calculus: $\int_a^b \frac{df}{dx}(x)\,dx = [f(x)]_a^b = f(b) - f(a)$.

Now recall that all states must be normalised, i.e. $\|\psi\| = \sqrt{\langle \psi | \psi \rangle} = 1$. For this integral to exist, all wavefunctions must vanish as x goes to either $+\infty$ or $-\infty$. Thus, $\psi^* \frac{\partial \zeta}{\partial x} - \frac{\partial \psi^*}{\partial x} \zeta$ is zero both as $x \to \infty$ and $x \to -\infty$, so our result vanishes. The difference between the inner products is zero, and so $\langle \psi | H \zeta \rangle = \langle H\psi | \zeta \rangle$ for all ψ and ζ. Therefore, $\boxed{H \text{ is self-adjoint}}$.